Ling 285
Human Language and Technology
Lecture 2

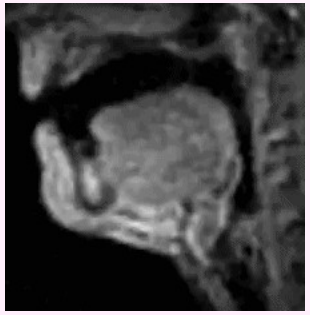
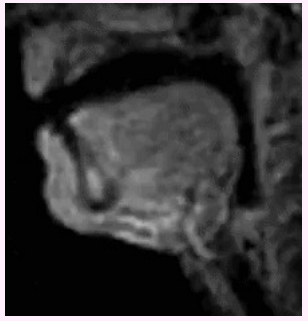
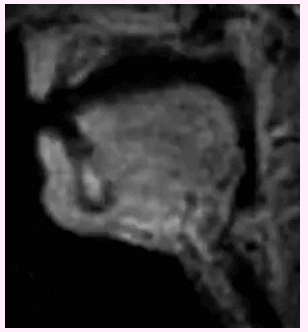
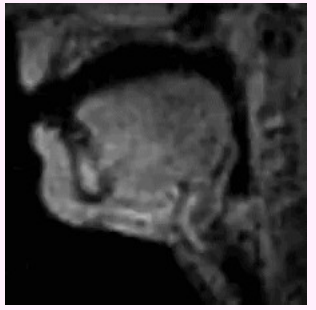
Khalil Iskarous
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Today

- Syllabus
- Phonetics
- What is AI?
- How does it know?
- How does it learn?
- (No discussion yet of Meadows)

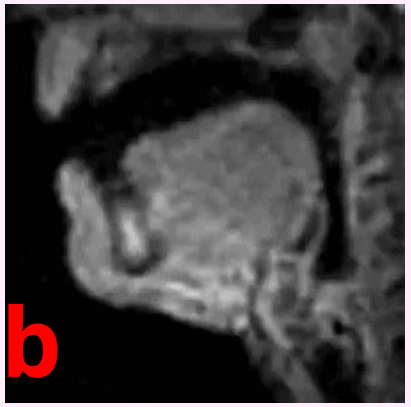
Fricatives

	Labial	Dental	Alveolar	Postalveolar
Voiceless	f	th	s	sh
Voiced	v	th	z	j

Stops

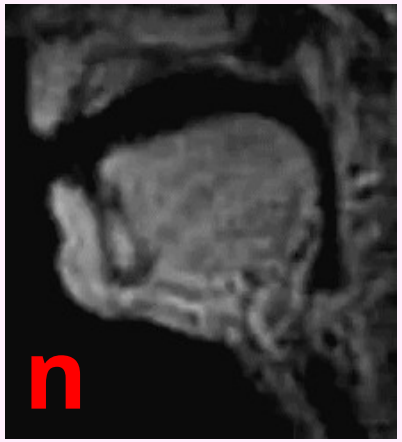








d





g

ng

Consonants 1

	Voice	Place of Articulation				
Manner of Articulation		Labial	Dental	Alveolar	Postalveolar	Velar
Fricatives	Voiceless	f	th	s	sh	
	Voiced	v	th	z	j	
Oral Stops (Plosives)	Voiceless	p		t		k
	Voiced	b		d		g
Nasal Stops	Voiced	m		n		ng

Do you see how much structure there is to your knowledge of English?????

What is AI?

- AI is a model of several animal psychological skills:
  - Knowing
  - Recognizing
  - Attending
  - Acting
  - Learning
- As you will learn later, and unlike what you will hear from many people: How AI models each of these things is at least slightly based on how humans do each of these things. AI is *NOT* some kind of technical innovation that came out of nowhere.

How does AI *know*?

Journalism Quarterly

DEVOTED TO RESEARCH STUDIES IN THE FIELD OF MASS COMMUNICATIONS

FALL 1953

"Cloze Procedure": A New Tool For Measuring Readability

BY WILSON L. TAYLOR*

Here is the first comprehensive statement of a research method and its theory which were introduced briefly during a workshop at the 1953 AEJ convention. Included are findings from three pilot studies and two experiments in which "cloze procedure" results are compared with those of two readability formulas.

"CLOZE PROCEDURE" IS A NEW PSYchological tool for measuring the effectiveness of communication. The method is straightforward; the data are easily quantifiable; the findings seem to stand up.

At the outset, this tool was looked on as a new approach to "readability." It was so used in three pilot studies and two experiments, the main findings of which are reported here.

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*The writer is particularly obligated to Prof. Charles E. Osgood, University of Illinois, and Melvin R. Marks, Personnel Research Section, A.G.O., Department of the Army, for instigating and assisting in the series of efforts that yielded the notion of "cloze procedure." Both are experimental psychologists. Among others who have advised, encouraged or otherwise aided are these of the University of Illinois: Prof. Lee J. Cronbach, educational psychologist and statistician; Dean Wilbur Schramm, Division of Communications; Prof. Charles E. Swanson, Institute of Communications Research, and George R. Klare, psychologist, both of whom have authored articles on readability; and several journalism teachers who lent their classes. Kalmer E. Stordahl and Clifford M. Christensen, until recently research associates of the Institute, also contributed.

First, the results of the new method were repeatedly shown to conform with the results of the Flesch and Dale-Chall devices for estimating readability. Then the scope broadened, and cloze procedure was pitted against those standard formulas.

If future research substantiates the results so far, this tool seems likely to have a variety of applications, both theoretical and practical, in other fields involving communication functions.

THE "CLOZE UNIT"

At the heart of the procedure is a functional unit of measurement tentatively dubbed a "cloze." It is pronounced like the verb "close" and is derived from "closure." The last term is one gestalt psychology applies to the human tendency to complete a familiar but not-quite-finished pattern—to "see" a broken circle as a whole one, for example, by mentally closing up the gaps.

One can complete the broken circle because its shape or pattern is so familiar that, although much of it actually is missing, it can be recognized anyway.

The same principle applies to language. Given "Chickens cackle and ——— quack," almost anyone can instantly supply "ducks." If that word really is the same as the one omitted, the person scores one cloze unit for correctly closing the gap in the language pattern.

Note that the sentence pattern is a complex one made up of many subpatterns. One must know not only the meanings (i.e., patterns of symbol-meaning relationships) and forms (patterns of letters) of all the five words, but also the meanings of given combinations of them—plus the fact that the sentence structure seems to demand a term parallel to "cackle" but associated with ducks instead of chickens. In other words, one must guess what the mutilated sentence means as a whole, then complete its pattern to fit that whole meaning.

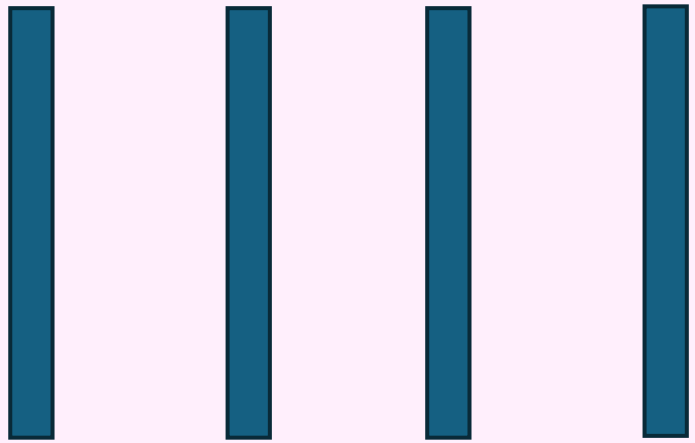
A cloze unit may be defined as: Any single occurrence of a successful attempt to reproduce accurately a part deleted from a "message" (any language product) by deciding, from the context that remains, what the missing part should be.

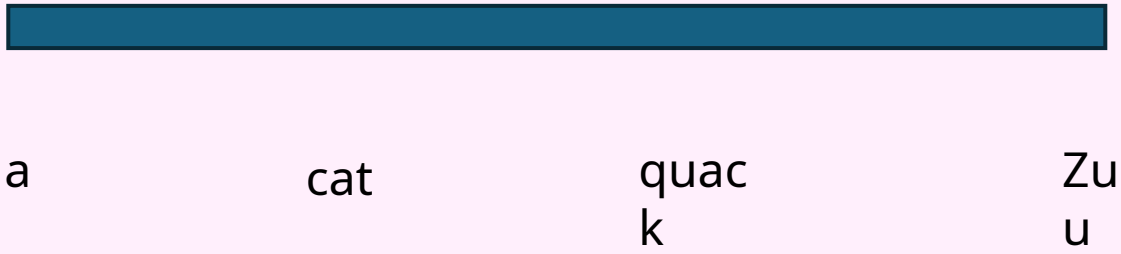
Cloze procedure may be defined as: A method of intercepting a message from a "transmitter" (writer or speaker), mutilating its language patterns by deleting parts, and so administering it to "receivers" (readers or listeners) that their attempts to make the patterns whole again potentially yield a considerable number of cloze units.

- *How* does AI *know*?
- By demonstrating the ability to *complete* or *cloze* sequences
- I can test your knowledge of a skill by asking you to complete some sequence
- AI: That is ALL what knowledge is: Completion!

GPT

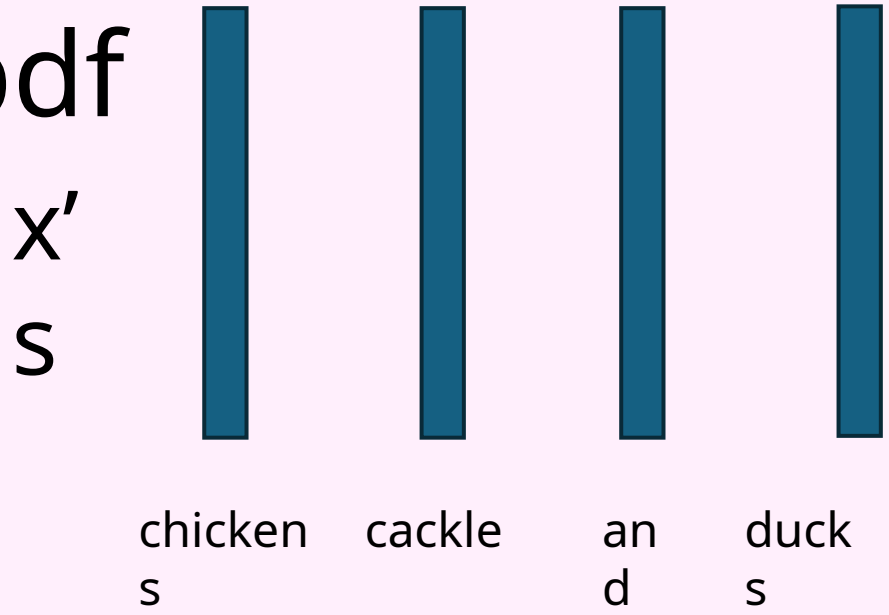
- How does it work?
- You give it a bunch of words
- What are words to a computer? A bunch of random numbers ☺
- Just make sure you don’t use the same numbers for each word.
- A bunch of number is called a *vector*. The vector could have 1024 numbers.
- So if the words are “chickens cackle and ducks ___”
- And the system, if it *knows English* will spit out a vector of 10000 numbers, each the probability that each word of English could be the next word.

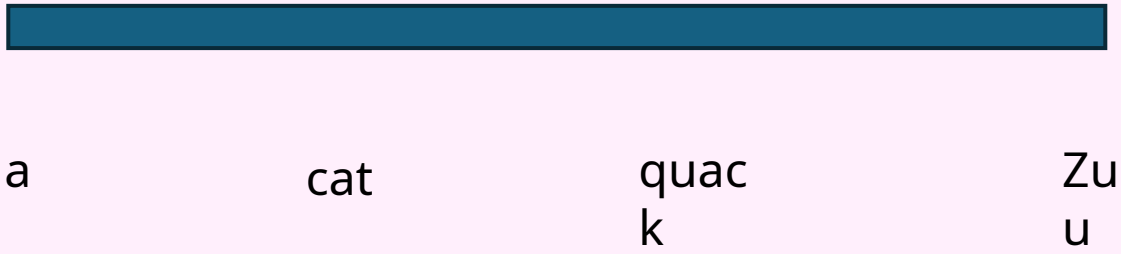




GPT: But how does it transform the input vectors to the pdf

- Well…
- Through recognizing by computing
- What is this *computing*?
- Mostly: +, - , × , ÷
- Yes, Grade School Arithmetic! Sorry!
- The knowledge by which the system transforms the x’s into the pdf is in a bunch of numbers called w’s , or *weights*
- Today’s AI systems have about a couple of trillions of these weights





Before you scream that these are two many to resemble human knowledge?

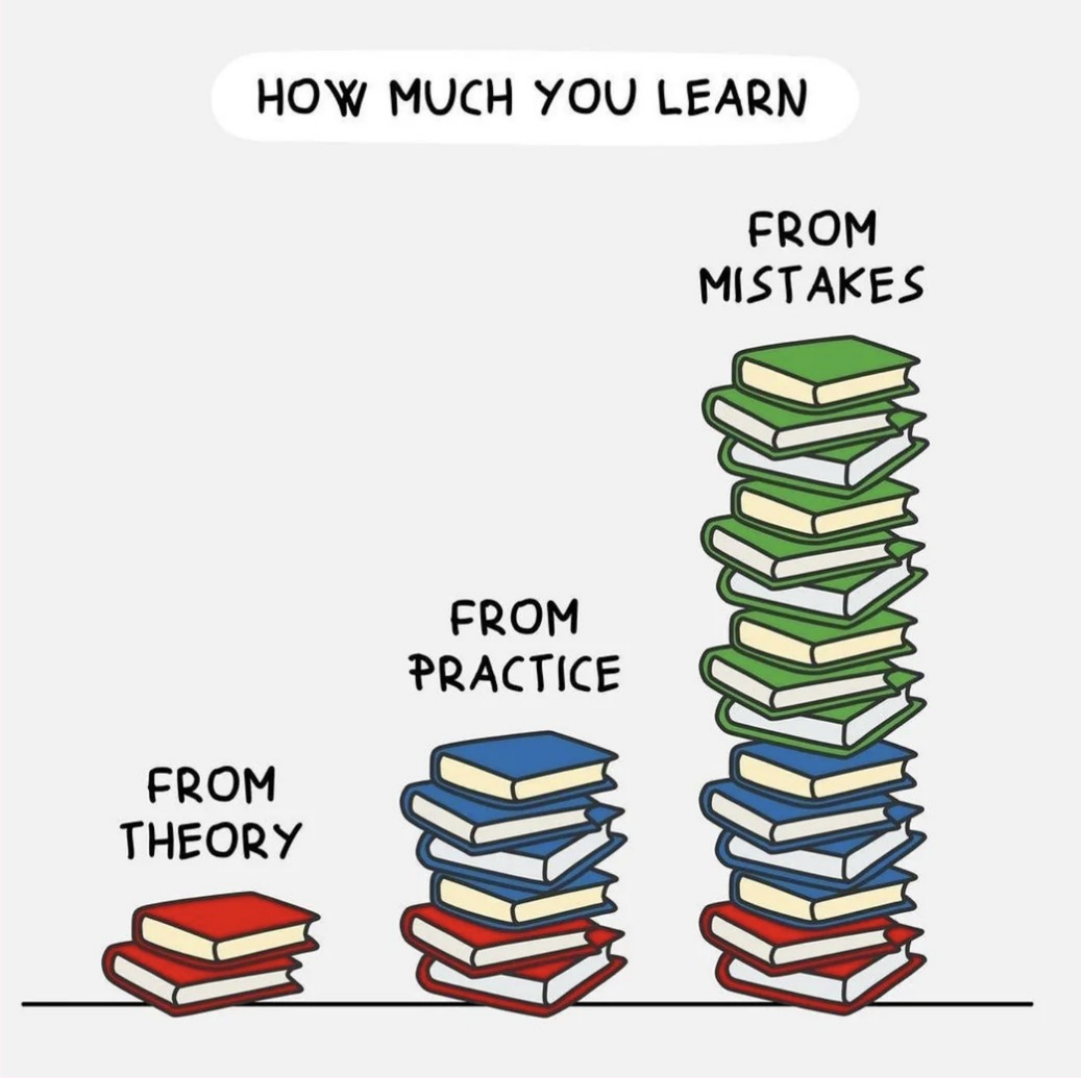
- How do you know that chickens cackle, but that ducks quack?
- This kind of knowledge is in the synapses between the neurons of your brain?
- How many synapses do you have?
- In or around 150 Trillion!

Still you may think: What????? Knowledge is Completion????? Come On!

- This is so nonsensical that Yann LeCun just raised a billion dollars out of this not making any sense.
- But why don’t you be the judge?
- If you are interacting with an AI system, do you think you are interacting with a system whose notion of knowledge is totally different from your own?
- Personally, for me, the answer is that whatever its notion of knowledge is, its not totally different from mine.
- Another term for Completion is Prediction. And that word sits better with people. We do predict, expect…

Now we will go to Learning

- One of the most central issues of AI is how it learns
- It learns by making mistakes
- We may not be that different…
- This is from a reddit motivational page
- We will now look at Stochastic Gradient Descent, the main algorithm of learning in AI!
- Sounds intimidating, but you can handle it, just stay focused
- We will learn with a very very easy example: Averaging



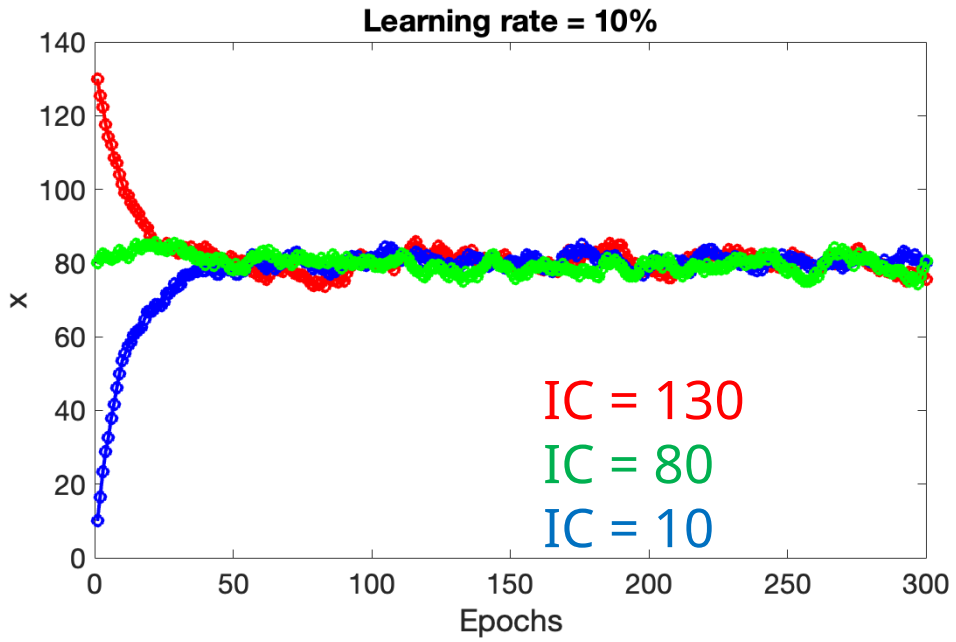
https://www.reddit.com/r/GetMotivated/comments/1072qzr/image_we_cant_avoid_making_mistakes_but_we_can/

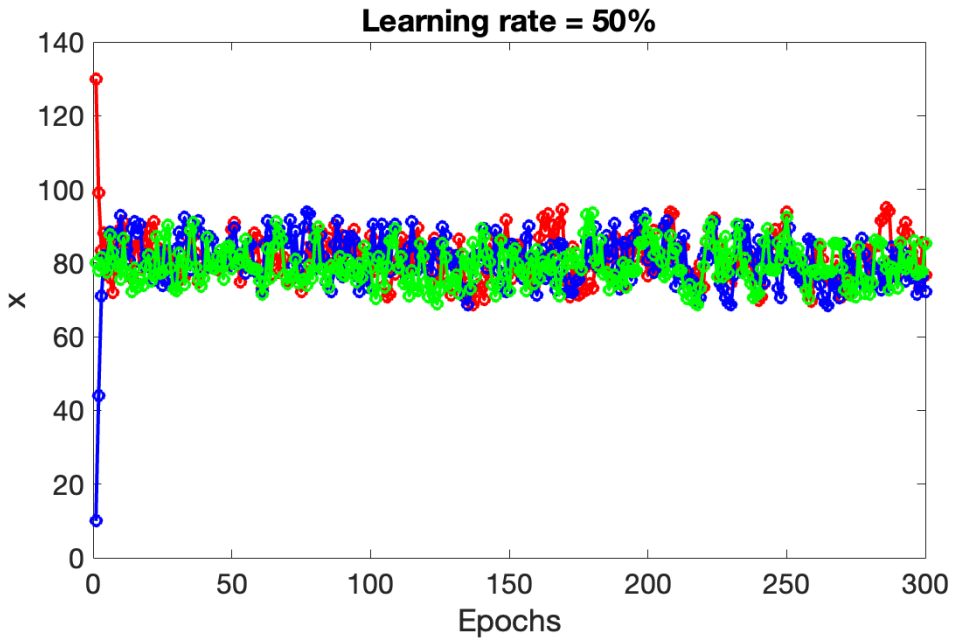
What's the big deal? Averaging is so easy

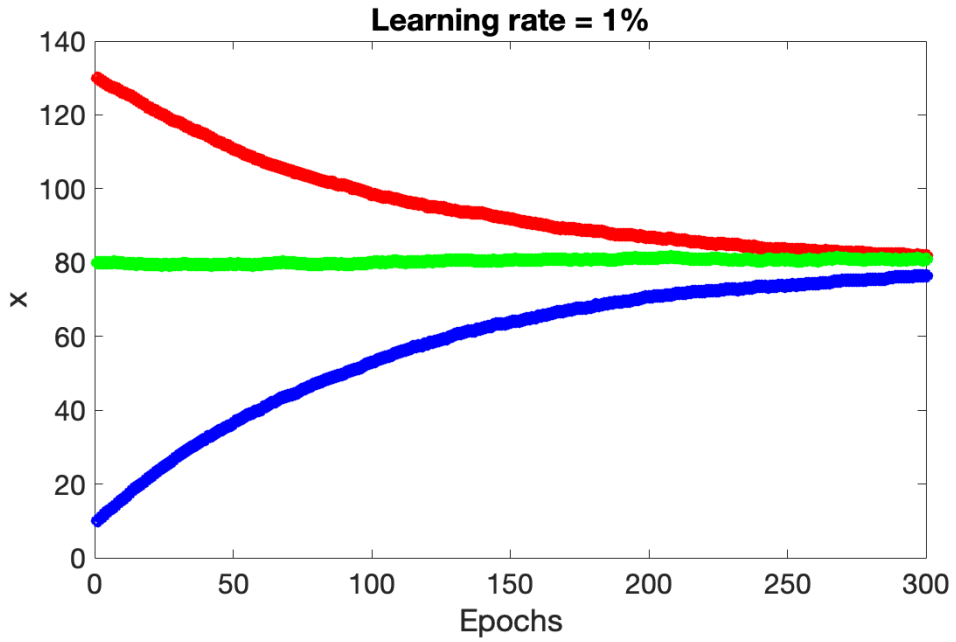
- What's the average of 10,12 and 14? $\frac{10+12+14}{3} = 12$
- But this won't work with more complex things like speech, language, biological molecules, scenes, etc., the things AI does…
- The averaging method we develop will work like this:
  - We have a list of numbers
  - We make some wild guess x about their average
  - We seek to improve x by exposure to the data. How?
    - We take one of the numbers in the list and examine it with respect to the current x
    - We now improve x by making it a bit closer to the number from the data we observed. This is observation-based error learning
    - Repeat with another number!

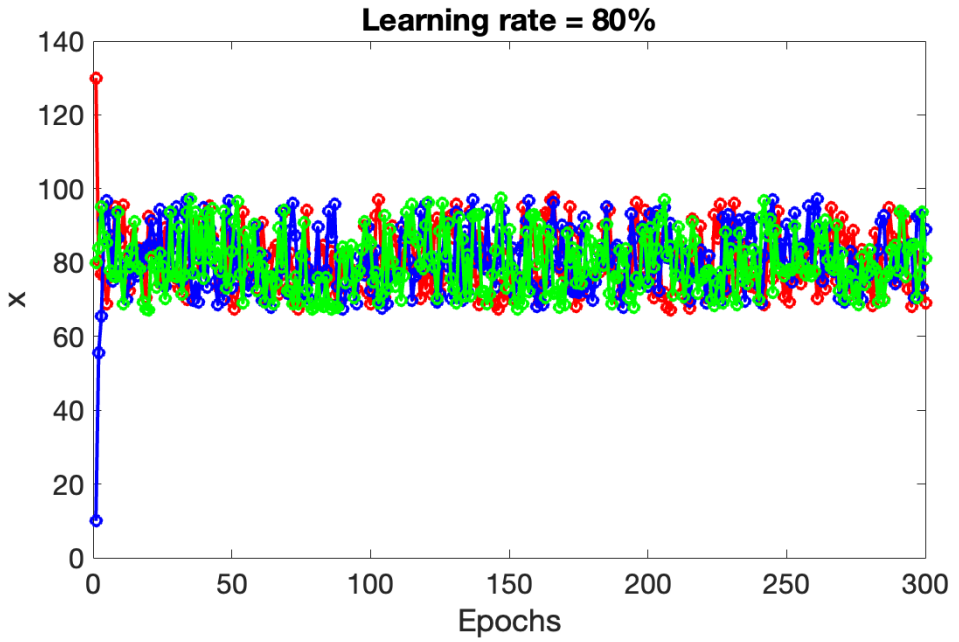
Stochastic Gradient Descent (SGD)

- What's the average, *the essence*, of: [78,67,98,85,76,68,98,72]?
  - 1) Pick any random number x as the estimate of average, and we will improve it by observation. We are at time 0 in the algorithm. Let's say the random number we pick is 130: x[0] = 130.
  - 2) Randomly pick a number d from the data, so we can see how far the data is from the estimate, *as the average should be close to the data*. Let's say we pick d[0] = 98.
  - 3) x[0] = 130 and d[0] = 98. The mistake we've made is Loss = 130 – 98 = 32.
  - 4) Now let's improve the estimate by reducing the discrepancy mistake = Loss by subtracting a bit of the Loss, let's say 10% from the current estimate.
    - x[1] = x[0] - 10% of 32
    - x[1] = 130 – 3.2 = 126.8
  - 5) Repeat: Now pick another number d from the data to improve x from x[1] to x[2].
- What have we done? 1) Made an intial estimate/guess, 2) Used data to improve the estimate.
- Stochastic (Random) Gradient (Loss-related) Descent (reduce)


Learning rate = 10%
x
Epochs
IC = 130
IC = 80
IC = 10


Learning rate = 50%
x
Epochs


Learning rate = 1%
x
Epochs


Learning rate = 80%
x
Epochs

Fundamental Learning Tradeoff

- $\gamma \uparrow$ Speed ↑ Accuracy ↓
- $\gamma \downarrow$ Speed ↓ Accuracy ↑
- ☹️
- Thousands of people for generations make their living by finding better way to *optimize* so that
  - Speed ↑
  - Accuracy ↑

